

LAB 1

Basic Network Setup & Connectivity

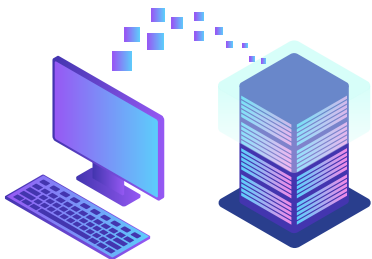
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CSIT491_05

LAB OVERVIEW

Description

The objective of this lab was to build a basic LAN in Cisco Packet Tracer by connecting a router, switch, and PCs. We configured the router's interface with an IP address, set up DHCP, and assigned static IP addresses to each PC. The lab focuses on verifying network configurations using `ipconfig` and `show ip interface brief`, then testing connectivity using `ping`.



Tasks

1

Identify and label each device in the network diagram

2

Assign static IP addresses & configure DHCP on router

3

Use `ping` to verify connectivity between all devices

4

Utilize `ipconfig` to confirm network settings on PCs

EQUIPMENT USED

01



1x Cisco ISR4331 Router

Connects the local network to other networks and acts as the default gateway for all devices.

02



1x Cisco 2960-24TT Switch

Connects multiple devices within the same local network so they can communicate with each other.

03



3x PC End Devices

Represent end-user computers used to test IP configuration and network connectivity.

04

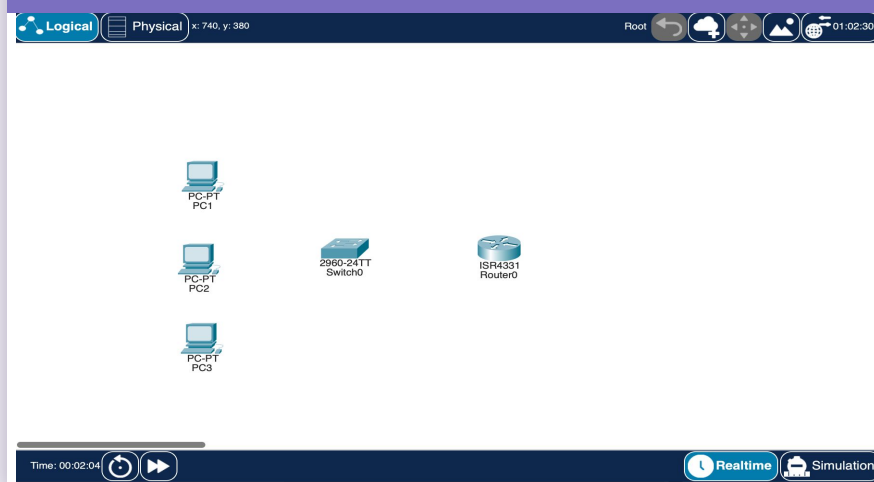


4x Straight-Through Cables

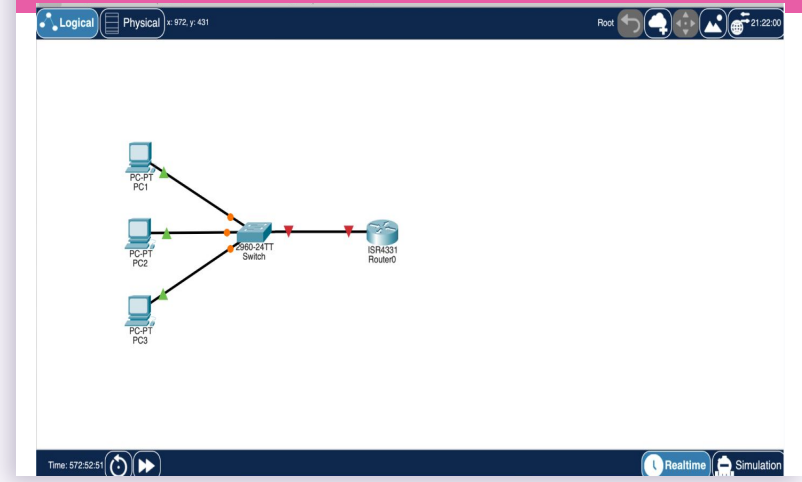
Used to physically connect each PC to the switch and the switch to the router.

NETWORK TOPOLOGY & CONNECTION

Step 1: Device Placement



Step 2: All Cables Connected



Devices were placed and physical connections were established. Three PCs were connected to the 2960 switch using Copper Straight-Through Ethernet cables, and the switch was connected to the ISR4331 router.

ROUTER CONFIGURATION

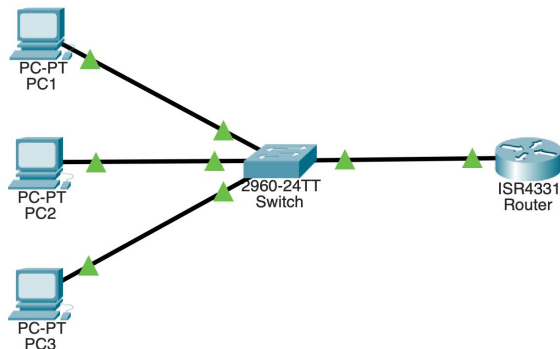
Router CLI Interface

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

Router(config-if)#exit
Router(config)#ip dhcp pool LAN
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
|
```



Interface IP Setup

Configured GigabitEthernet0/0/0 with IP address 192.168.1.1 and subnet mask 255.255.255.0. Enabled the interface using the no shutdown command.

No Shutdown



Enabled the interface using the no shutdown command to bring it online

DHCP Pool

Created a DHCP pool named LAN, defined network 192.168.1.0/24, and set the default router to 192.168.1.1 for automatic IP assignment.

PC STATIC IP CONFIGURATION

Each PC was manually configured with a static IPv4 address within the 192.168.1.0/24 network using a subnet mask of 255.255.255.0. The default gateway was set to 192.168.1.1, and unique IP addresses were assigned to ensure proper communication and avoid address conflicts.

IP Assignments

DEVICE	IP ADDRESS	GATEWAY
PC1	192.168.1.10	192.168.1.1
PC2	192.168.1.11	192.168.1.1
PC3	192.168.1.12	192.168.1.1

Subnet Mask: 255.255.255.0

PC1 Configuration Panel

PC1

PhysicalConfigDesktopProgrammingAttributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

☐ DHCP

☒ Static

IPv4 Address192.168.1.10

Subnet Mask255.255.255.0

Default Gateway192.168.1.1

DNS Server0.0.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address/

Link Local AddressFE80::230:F2FF:FE77:D07A

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

AuthenticationMD5

Username

Password

Top

NETWORK VERIFICATION

ipconfig run on each PC to confirm IP address, subnet mask, and default gateway.

PC1 — .10

```
PC1
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ipconfig

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix...: FE80::20A:F3FF:FE3A:98D2
    Link-local IPv6 Address...: FE80::20A:F3FF:FE3A:98D2
    IPv4 Address...: 192.168.1.11
    IPv4 Subnet Mask...: 255.255.255.0
    Default Gateway...: 192.168.1.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address...:
    IPv4 Address...: 0.0.0.0
    IPv4 Subnet Mask...: 0.0.0.0
    Default Gateway...: 0.0.0.0
```

PC2 — .11

```
PC1
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ipconfig

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix...: FE80::20A:F3FF:FE3A:98D2
    Link-local IPv6 Address...: FE80::20A:F3FF:FE3A:98D2
    IPv4 Address...: 192.168.1.11
    IPv4 Subnet Mask...: 255.255.255.0
    Default Gateway...: 192.168.1.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address...:
    IPv4 Address...: 0.0.0.0
    IPv4 Subnet Mask...: 0.0.0.0
    Default Gateway...: 0.0.0.0
```

PC3 — .12

```
PC3
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ipconfig

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix...: FE80::20A:F3FF:FE3A:98D2
    Link-local IPv6 Address...: FE80::20A:F3FF:FE3A:98D2
    IPv4 Address...: 192.168.1.12
    IPv4 Subnet Mask...: 255.255.255.0
    Default Gateway...: 192.168.1.1

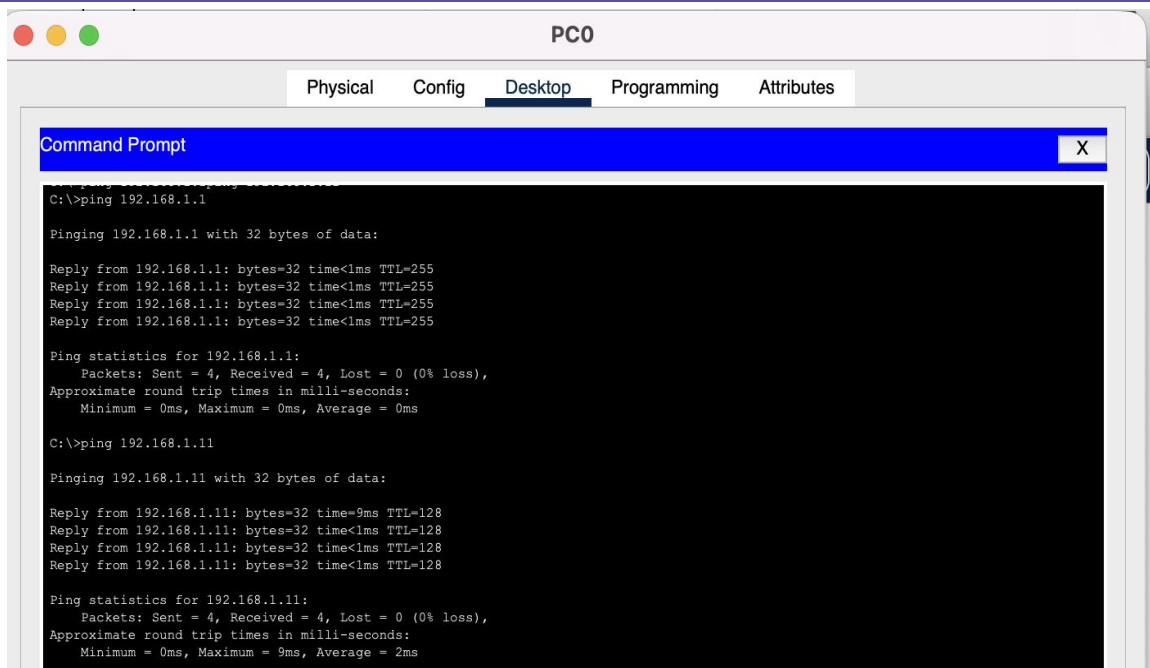
Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address...:
    IPv4 Address...: 0.0.0.0
    IPv4 Subnet Mask...: 0.0.0.0
    Default Gateway...: 0.0.0.0
```

The ipconfig command was used on each PC to verify that the correct IP address, subnet mask, and default gateway were assigned. The outputs confirm that all three devices are configured within the 192.168.1.0/24 network and are properly set to communicate through the router (192.168.1.1).

CONNECTIVITY TESTING

PC0 – ping 192.168.1.1 / .11 / .12



The screenshot shows a virtual PC window titled 'PC0' with tabs for Physical, Config, Desktop (selected), Programming, and Attributes. A Command Prompt window is open, displaying the results of two ping commands. The first command is 'ping 192.168.1.1', which shows four successful replies with 0% packet loss. The second command is 'ping 192.168.1.11', which also shows four successful replies with 0% packet loss.

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time=9ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 9ms, Average = 2ms
```



Packets
Received



Packet Loss



Hosts Reached

CHALLENGES FACED & SOLUTIONS

CHALLENGE

Incorrect IP Address Configuration

SOLUTION

Used ipconfig to validate settings and confirmed correct IP address, subnet mask, and gateway were properly entered.

CHALLENGE

No Connectivity Between Devices

SOLUTION

Used ping to test connectivity. Checked cable connection status and confirmed the router interface was active.

CHALLENGE

DHCP Not Assigning IP Addresses

SOLUTION

Verified DHCP configuration on the router and restarted the client device to trigger a fresh address request.

LESSONS LEARNED

01

Layering Logic

We noticed that you can have a perfect physical link when the light is green through the switch. If the IP logic is flawed the ping will fail.

02

Default Gateway

In the router's "inner" interface we picked up on the importance of gateway addresses being required in order for the computer to talk to its neighboring PC.

03

Verification Cruciality

The ipconfig allowed us to see the settings that the computer had taken from the policy we directed and using the ping we learned that it directed us on how the round-trip path was clear and to ask ourselves whether it was a problem with the cable, IP, or the Firewall blocking us.

REAL-WORLD APPLICATIONS



Hospitals

Making sure medical devices along with record systems of the patients can stay connected and reachable throughout



Academies & Universities

Manage lab computers through centralized DHCP services, and regularly test connectivity.



Home Networks

Routers capable of connecting automatically to ip addresses that are assigned while users troubleshoot using *ping* and *ipconfig*.



Retail Stores

Cash registers and payment terminals have to be connected to the store's network in order for credit cards to process transactions

